A Computer-Vision-Based Bikeability Score

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***Abstract*—**

***Index Terms*—Bikeability, computer vision, zero-shot classification, vision-language models, CLIP, surface classification, object detection, GPX.**

I. Introduction

II. Research Approach and Methodology

III. Objectives and Constraints

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IV. ROAD-SURFACE CLASSIFICATION

The ridden surface strongly affects cycling comfort and safety: a paved cycleway is effortless, whereas loose gravel or bare soil is strenuous and, on a road bike, hazardous. Ground detection assigns each sampled frame a single surface class and supplies the surface term of the score. Frames come from a handlebar-mounted camera at one frame every five seconds; as each is dominated by one surface, the task is single-label classification, not segmentation.

A. Task Definition and Model Selection

Two model families can solve this task: a classifier *trained on our own labels*, or a *zero-shot* vision-language model (VLM). We avoid a self-trained classifier, which would need a large, balanced, manually labeled training set and re-training on every change of the class definitions. A zero-shot VLM of the CLIP family [1] instead classifies an image by its similarity to natural-language class descriptions, so the taxonomy is adapted by editing text alone.

B. Surface Classes and Prompts

We distinguish four cycling-relevant classes: **Cycleway** (a dedicated paved bike path), **Road** (a paved road for motor traffic), **Gravel** (loose crushed stone), and **Unpaved** (soil, mud, or field tracks). Each class is described by several English prompts whose normalized embeddings are averaged (*prompt ensembling*). As Cycleway and Road share the same smooth asphalt, the prompts encode the *discriminating context* (a narrow separated bicycle lane versus a wide multi-lane road) to separate the two embeddings. A frame is assigned to the class of highest cosine similarity.

C. Evaluation Protocol

We compare nine open CLIP-family models of light to medium size through the unified `open_clip` API [2]: OpenAI CLIP, OpenCLIP (LAION-2B, DataComp-XL) [3], SigLIP [4], MobileCLIP [5], and EVA-CLIP [6]; both *accuracy* and *latency* matter for on-bike use. For the quantitative evaluation we manually labeled more than 1,700 frames from self-recorded rides, excluding frames without a clear surface. As the classes are imbalanced, we report accuracy, balanced accuracy and macro-F1 with 95% bootstrap confidence intervals and McNemar tests; latency is measured at batch size 1 with warm-up runs (ms/frame and FPS).

D. Results

Table I lists the outcome, sorted by macro-F1. The best model, **OpenCLIP ViT-B/32 (LAION-2B)**, reaches macro-F1 0.58 at only 115 ms/frame, the best accuracy-latency trade-off. Two points stand out. First, the highest plain *accuracy* (0.78, CLIP ViT-B/16) does *not* give the highest macro-F1: accuracy rewards the majority classes, whereas macro-F1 weights all four surfaces equally and is the more faithful measure on imbalanced data. Second, the 95% bootstrap accuracy intervals of the leading models overlap, so their differences are *not* statistically significant; only CLIP ViT-L/14 (428 M) falls clearly below them; the largest model is both slowest and weakest (0.42 macro-F1), confirming that capacity alone does not help.

Table I
ZERO-SHOT SURFACE-CLASSIFICATION BENCHMARK ON THE SELF-LABELED TEST SET (>1,700 FRAMES), SORTED BY MACRO-F1. BACC: BALANCED ACCURACY; LATENCY AT BATCH SIZE 1.

Model	P[M]	Acc	BAcc	mF1	ms	FPS
OpenCLIP B/32 (LAION-2B)	151	0.75	**0.72**	**0.58**	115	8.7
SigLIP B/16 (WebLI)	203	0.75	0.62	0.57	259	3.9
CLIP B/16 (OpenAI)	150	**0.78**	0.62	0.57	261	3.8
MobileCLIP-S2	99	0.73	0.70	0.56	304	3.3
CLIP B/32 (OpenAI)	151	0.70	0.60	0.54	93	10.7
EVA02-B/16	150	0.76	0.62	0.54	304	3.3
OpenCLIP B/32 (DataComp-XL)	151	0.75	0.65	0.52	125	8.0
CLIP L/14 (OpenAI)	428	0.64	0.50	0.42	929	1.1

Broken down per class, the best model scores F1 0.80 (Cycleway) and 0.73 (Road) but only 0.45 (Gravel) and 0.33 (Unpaved); as macro-F1 averages the four equally, the two strong classes are pulled down by the two weak ones (Fig. 1). Crucially, the weakness is one of *precision*, not recall: Unpaved is still recalled well (0.77), yet its precision collapses to 0.21 because, with only 13 true frames, a few false positives bleeding in from the majority classes suffice to ruin it; Gravel (51 frames) behaves the same way, while the residual Cycleway/Road confusion (both smooth asphalt) is comparatively minor. A macro-F1 of 0.58 is thus by design strict on such tiny classes; the accuracy (0.75) and balanced accuracy (0.72, the highest of all models) are more flattering yet still fair, and for a zero-shot classifier on strongly imbalanced data the result is solid. We therefore adopt OpenCLIP ViT-B/32 (LAION-2B) as the ground detector; as all models share the same pipeline, the differences reflect architecture and training data (latencies are hardware-dependent).

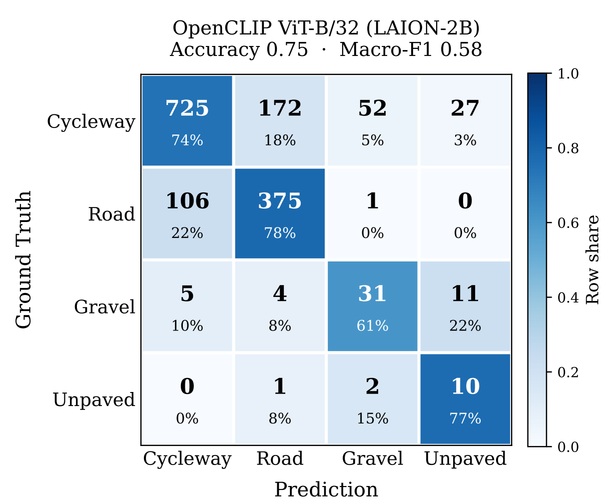

Figure 1. Row-normalized confusion matrix of the best model (OpenCLIP ViT-B/32, LAION-2B) on the labeled test set. Each cell shows the count and the row share; the diagonal is correct. Errors concentrate on the Cycleway/Road pair and the rare Gravel/Unpaved classes.

V. Object Detection

VI. ENVIRONMENT DETECTION: ZERO-SHOT VS. FINE-TUNED SEMANTIC SEGMENTATION

Riding through different environments shapes the experience: green *vegetation* and *water* make a ride pleasant, too much *city* and traffic make it strenuous. Environment detection is a *multi-label whole-image* task (a frame may show several classes or none) and supplies the *largest* score weight ($w_e = 0.45$, Eq. (2)).

A. Task and Model Selection

Task: find the best model to classify the environment on our own test frames, captured with a pod-mounted camera on the bike. **Comparison:** training-free *zero-shot CLIP* [1], [7] vs. a *fine-tuned SegFormer* [8].

B. Model Comparison

CLIP (zero-shot) scores each frame against a per-class prompt pair in a shared embedding space, yielding *one present-probability per class* – no *where* or *how much* – thresholded into a label, and is tuned only via prompts and thresholds. **SegFormer (fine-tuned)** instead labels *every pixel* and reports each class's *area fraction* (pixels ÷ total), the *how much* the score consumes [9] (Eq. (4)); only it yields those fractions and a pixel-level mIoU, at the cost of training.

C. Zero-Shot CLIP: Optimization

Two levers dominated (Table II): *per-class threshold calibration* on a leakage-free validation split (+0.095 macro-F1) and *prompt ensembling* [1] over 6–8 templates per class (+0.044); larger capacity (ViT-L/14) did *not* help. Adding the water rides in the final test (C7) then exposed CLIP's apparent water strength as a small-validation artifact.

Table II
CLIP ZERO-SHOT OPTIMIZATION: CHRONOLOGICAL OWN-TEST MACRO-F1. C1–C6 ARE PRE-WATER; C7 ADDS THE WATER RIDES (BASIS CHANGE), WHERE CLIP'S WATER PROVES AN ARTIFACT. FINAL: VIT-B/32, PROMPT ENSEMBLING.

#	Optimization step	Macro-F1
C1	Aligned prompt pairs, global threshold	0.547
C2	Ride-disjoint val/test split	0.632
C3	Per-class threshold calibration	0.727
C4	Prompt ensembling, 6–8 templates	0.771
C5	Larger backbone ViT-L/14 (rejected)	0.756
C6	3-class taxonomy (veg. merge)	**0.846**
C7	Water-inclusive final test	0.649

D. Fine-Tuned SegFormer: Optimization

Tuning ran in two phases (Table III). *(i) Training* on Mapillary [10] pixel masks, by **mIoU**: the decisive lever was *data* – balanced oversampling plus 258 ADE20K water scenes [11] lifted water IoU 0→0.70, and an ADE20K base [12] with an "other" class raised it further. *(ii) Evaluation* on held-out own frames, by **macro-F1**, ending on the unseen water-inclusive test (0.704); ablation S10 values fine-tuning at +0.027.

Table III
SEGFORMER OPTIMIZATION IN TWO PHASES: *training* ON MAPILLARY PIXEL MASKS (MIOU, DEV SET), THEN *evaluation* ON OWN FRAMES (MACRO-F1), ENDING ON THE UNSEEN WATER-INCLUSIVE TEST. FINAL: B0 ADE20K BASE, 3 CLASSES + "OTHER".

#	Optimization step	$\mathrm{mIoU}_{\mathrm{tr}}$	$\mathrm{F1}_{\mathrm{eval}}$
S1–2	Cityscapes-B0, random Mapillary; LR tuning	0.62	–
S3–4	Balanced + water aug + ADE water	0.82	–
S5	Base swap → ADE20K (water prior)	**0.857**	–
S7	First own eval + val thresholds	–	0.636
S8	Explicit "other" background class	–	0.707
S9	3-class retrain (pre-water)	–	0.874
S10	Fine-tuning ablation (no-FT)	–	0.847
S11	Threshold grid, water-incl. final	–	**0.704**

E. Metrics

Two metrics on the **1,210**-frame own-test set (ride-disjoint from tuning; 983/40/307 veg/water/city positives). **Macro-F1**: per-class F1 (precision–recall harmonic mean) averaged equally over classes, so rare *water* counts fully. **Exact-match**: share of frames whose entire three-class label is correct at once.

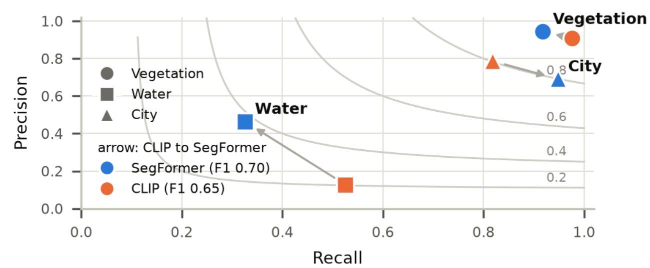

Figure 2. Precision–recall operating points with iso-F1 contours (own-test, 1,210 frames); arrows run CLIP→SegFormer. The isolated water pair (bottom-left) decides it: SegFormer trades recall for precision to reach a higher F1.

F. Results and Decision

Results. SegFormer wins overall: macro-F1 0.704 vs. 0.649 and exact-match 0.755 vs. 0.691. The models tie on the common classes (vegetation 0.930/0.941, city 0.799/0.802), so the rare *water* class (3% of frames) decides it: SegFormer F1 0.382 vs. 0.206. Figure 2 shows two things: *(1)* the decision lives in the bottom-left – vegetation and city cluster on the same high-F1 band, only the water points are far apart; *(2)* SegFormer wins water by *precision, not recall*: CLIP reaches higher water recall (0.53 vs. 0.33) only by firing indiscriminately at precision 0.13, whereas SegFormer holds precision 0.46 for a higher F1.

Decision. We deploy the fine-tuned SegFormer: it wins the deciding class with trustworthy precision, ties CLIP elsewhere, is 40× smaller (3.7 M vs. 151 M), and feeds the score area fractions directly – which CLIP cannot. *Water stays the known limitation*: recall (∼67% of true-water frames missed) is capped by data scarcity, fixable only with more in-domain water.

VII. Computation of the Bikeability Score

The bikeability score combines the three components (surface from ground-detection, environment from environment-detection, and objects from object-detection) into a single value in the interval $[0, 100]$. Every detector emits a fixed-length *vector*; each vector is first mapped to a *sub-score* in $[0, 1]$, and the three sub-scores are then combined by a weighted mean.

A. Detector Outputs as Vectors

The three detectors deliberately use different but fixed output representations, each aligned element by element with a stored value vector:

- **Ground**: a one-hot vector $\mathbf{g} \in \{0, 1\}^4$ over the classes (Cycleway, Road, Gravel, Unpaved); exactly one entry is 1.
- **Environment**: a vector of area fractions $\mathbf{f} \in [0, 1]^3$ over (vegetation, water, city).
- **Object**: a vector of counts $\mathbf{n} \in \mathbb{Z}^3_{\geq 0}$ over (bicycle, car, traffic light).

B. Overall Formula

For each evaluated frame, the score A results from the weighted mean of the three sub-scores:

$$A = 100 \cdot \Big(w_g\, S_{\text{ground}} + w_e\, S_{\text{env}} + w_o\, S_{\text{object}}\Big), \tag{1}$$

with the weights

$$w_g = 0.20, \quad w_e = 0.45, \quad w_o = 0.35. \tag{2}$$

The weights are heuristic and follow two considerations. Environment and object exposure jointly capture how pleasant and how safe a stretch feels, the factors a cyclist perceives most directly, and therefore receive the larger share ($w_e + w_o = 0.80$). The surface receives the smallest weight ($w_g = 0.20$): on typical routes it varies little, and ground detection is also the least reliable component (macro-F1 0.58), so a smaller weight limits the influence of its errors. Since all sub-scores lie in $[0, 1]$ and the weights sum to 1, the score is bounded to $[0, 100]$. In practice the lower extreme 0 is never reached, as $S_{\text{ground}} \geq 0.1$ and $S_{\text{object}} = \exp(-\max(\rho, 0)) > 0$; the upper value 100, however, does occur whenever all three sub-scores equal 1 simultaneously, e.g. on a paved cycleway ($S_{\text{ground}} = 1$) through vegetation or along water ($S_{\text{env}} = 1$) with no motorised traffic ($S_{\text{object}} = 1$).

C. Ground Sub-Score

The ground sub-score is the inner product of the one-hot vector $\mathbf{g}$ with the surface quality vector $\mathbf{q}^g = (1.0, 0.7, 0.3, 0.1)$ (Table IV):

$$S_{\text{ground}} = \langle \mathbf{g}, \mathbf{q}^g \rangle = \sum_k g_k\, q^g_k. \tag{3}$$

Because $\mathbf{g}$ is one-hot, this simply selects the quality of the detected surface class.

D. Environment Sub-Score

The environment sub-score is the fraction-weighted mean of the environment quality vector $\mathbf{q}^e = (1.0, 1.0, 0.4)$:

$$S_{\text{env}} = \begin{cases} \dfrac{\langle \mathbf{f}, \mathbf{q}^e \rangle}{\sum_k f_k}, & \text{if } \sum_k f_k > 0, \\ 0.5, & \text{otherwise.} \end{cases} \tag{4}$$

The normalization by $\sum_k f_k$ keeps the sub-score in $[0, 1]$ even when the detected fractions do not cover the whole frame; a neutral value of 0.5 is used if no environment class is detected.

E. Object Sub-Score

The object sub-score maps the counts $\mathbf{n}$ through a *saturation function* using the penalty vector $\boldsymbol{\lambda} = (-0.2, 1.0, 0.3)$:

$$S_{\text{object}} = \exp\Big(-\max\left(\rho, 0\right)\Big), \qquad \rho = \langle \mathbf{n}, \boldsymbol{\lambda} \rangle, \tag{5}$$

where ρ is evaluated independently per frame. A positive factor for cars (+1.0) and traffic lights (+0.3) makes ρ positive and thus lowers the sub-score, whereas bicycles carry a *negative* factor (−0.2). The clamp $\max(\rho, 0)$ makes the score asymmetric: a frame is never rewarded *above* the neutral value 1.0, so bicycles do not inflate the score on their own; they only *offset* the penalty of nearby cars or traffic lights. This encodes the intuition that good cycling conditions are the default and only motorised traffic degrades them. The exponential form additionally caps the influence of outliers, so a single crowded frame cannot dominate the overall score.

Table IV
Class value vectors for the three sub-scores: surface quality $\mathbf{q}^g$, environment quality $\mathbf{q}^e$, and object weight $\boldsymbol{\lambda}$ (heuristic; a negative value acts as a bonus).

Component	Class	Value
Ground $\mathbf{q}^g$	Cycleway	1.0
	Road	0.7
	Gravel	0.3
	Unpaved	0.1
Environment $\mathbf{q}^e$	Vegetation	1.0
	Water	1.0
	City	0.4
Object weight $\boldsymbol{\lambda}$	Bicycle	−0.2
	Car	+1.0
	Traffic light	+0.3

F. GPX Mapping and Visualization

Because the camera clock is unreliable, the frames are aligned to the GPS track by a single *manual sync point* (by default: recording start = GPX track start). Each frame is then matched to its nearest GPX track point in time (a nearest-neighbor join within a 3 s tolerance: large enough to bridge gaps in the roughly 1 Hz GPS log, yet below the 5 s frame spacing so that each frame maps to a distinct track point), which assigns a latitude/longitude to every score. The *average* bikeability score of the route is the mean $\bar{A} = \frac{1}{M} \sum_{i=1}^{M} A_i$ over all M evaluated frames. Finally, the route is drawn as a sequence of short segments, each colored by the local score on a red–orange–yellow–green scale (Fig. 3). For the example ride shown, the mean score is $\bar{A} \approx 85.7/100$: the track is predominantly green (good cycling conditions) with a few yellow/orange sections.

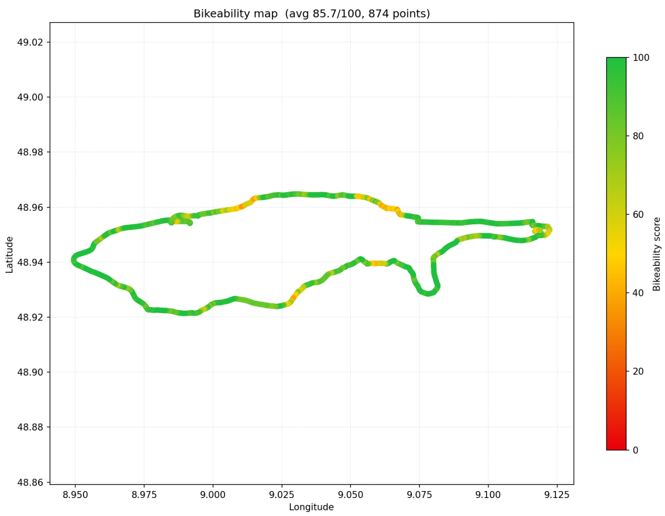

Figure 3. Bikeability score colored along the GPX route (red = low, green = high). Per-frame scores are matched to the nearest GPS track point via a manual sync point; the example ride has a mean score of $\bar{A} \approx 85.7/100$.

VIII. Discussion

IX. Conclusion

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